



Project ID: Eng575

Category: Engineering

Authors:

Author 1: Spulber George, 9th grade, "Ferdinand I" National High School, Bacau, Romania

Author 2: Mutica Tudor, 10th grade, "Ferdinand I" National High School, Bacau, Romania

Supervisor: Întuneric Ana, computer science teacher, "Ferdinand I" National High School, Bacau, Romania

Keywords: drone, emergency, face-recognition, firefighter, paramedic

The FDP Drone (Firefighter, Doctor, Police Drone) project is a cutting-edge solution to address the inefficiencies and delays of emergency response in urban areas. The drone has evolved into a device that can intervene ahead of police, fire, and medical crews to increase the chance of survival for individuals in danger.

The primary objective of the FDP Drone project is to increase the chances of survival for people in dangerous situations. It can help rescue crews by handing out first aid kits, communicating with victims by audio-visual communication, and transmitting information to the pilot/dispatch in real time. The drone can be controlled from anywhere in the world, which allows it to avoid road traffic and quickly reach the site of an emergency. In addition, the device is equipped with technology that can extinguish low-intensity fires and chase away aggressive animals using ultrasonic waves (with systems under development).

The FDP Drone project is a result of applying computer science, mathematics, mechanics, electronics, and programming skills. The drone is a hexacopter based on a carbon fiber frame, equipped with six GARTT ML 5008 motors that can rotate at up to 8800 RPM, and six 15-inch propellers. To control the motors, the drone uses a Pixhawk 2.4.8 flight controller based on an STM32 microprocessor. The flight controller's role is to take care of the control of the engines according to the data received from the user, with the help of the sensors implemented in it.

The FDP Drone project uses two cameras, a radio camera, and a web camera to transmit live images over short distances and observe images over the internet, respectively. The drone also has facial recognition capabilities using the OpenCV library and can detect objects in images and video streams using the YOLO library. For audio communication with people in danger or those who can offer help, the drone uses a radio station.

The FDP Drone project also aims to stimulate and encourage the involvement of individuals in developing technical projects and solving complex tasks in a creative way. The project seeks to foster a sense of team spirit, share experiences and knowledge, and promote the development of projects with a positive social impact. The project's educational framework combines programming, design, and mathematical skills, following the STEAM concept, which integrates knowledge into a coherent learning paradigm based on real-world applications.

Overall, the FDP Drone project represents a significant breakthrough in emergency response technology, with the potential to save countless lives. Its multidisciplinary approach and educational framework make it an excellent example of how technical skills can be combined with social responsibility to create a positive impact.